

Leo Zhang

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PROFESSIONAL SUMMARY

Master of Computer and Information Sciences (**First Class Honours**) from AUT. AI Software Engineer — AI engineer with deep experience in **computer vision**, machine learning, image processing, **LLM** orchestration and production-grade GenAI application development. Delivered ChatClothes, a multimodal virtual try-on system integrating diffusion models, computer vision and local LLM workflows — from concept through evaluation, deployment and publication. Involved extensive image dataset work, **model** training and testing across multiple CV projects. Optimised AI inference for ARM edge hardware (Raspberry Pi, Android) with offline-capable, on-device prediction. Delivery focus balances **model** quality with production readiness.

CORE COMPETENCIES

AI / ML: PyTorch, OpenCV, Image Processing, **Model** Training & Testing

Infrastructure: Python, FastAPI, Docker, Kubernetes

Data & Deployment: ETL, **Data** Processing, SQLite, MySQL

Product Integration: API Integration, User Workflow Design, Offline-capable Systems

AI-Assisted Development: Cursor, GitHub Copilot, Claude Code, OpenCode

EXPERIENCE

Auckland University of Technology — *Auckland, New Zealand*

2024 – 2026

AI Research Engineer

- Designed and delivered ChatClothes, a digital multimodal AI platform integrating computer vision, diffusion models, and LLM-powered workflows with RAG, vector search, and prompt grounding. Conducted extensive image dataset work, **model** training and testing across multiple CV projects. Provided technical leadership and owned the product end-to-end as **cloud-native AI solutions**, from system design and development through evaluation, deployment, and user validation.
- Deployed and optimized AI models on edge devices including Raspberry Pi and Android phones, running lightweight LLMs and vision models on resource-constrained ARM hardware with offline inference, image processing, and on-device prediction. Validated **model** outputs through ground truth testing and evaluation.

Tech: Python, PyTorch, Ollama, Docker, Linux, AWS, Azure, Git, Raspberry Pi, Android

Chunxiao Technology Co., Ltd. — *China*

2013 – 2024

*Technology company (est. 2013, ~40 employees) delivering software, hardware, and system integration across streaming media, enterprise platforms, IoT, and industrial **automation**.*

Senior Software Engineer | 2018 - 2024

- Smart Factory IoT Platform: architected manufacturing platform across 10+ factory sites with 6-person cross-functional team. Integrated RFID readers, barcode scanners, electronic scales (RS232/RS485 serial bridge to WebSocket), conveyors, and Android workshop terminals with Spring **Cloud** microservices. Event-driven architecture via ActiveMQ/Kafka, Docker deployment, 30%+ efficiency gain.
- IoT Device Management Platform: built Android device discovery/binding/remote control app and gateway firmware for smart switches and Zigbee/WiFi gateways. Protocol handling, MQTT **cloud** uplink, Spring **Cloud** backend API, and Vue.js management dashboard.

- Boobit Crypto Trading App: sole Android developer for financial cryptocurrency trading app. Real-time market **data** via WebSocket, order book rendering, candlestick charting, encrypted local wallet storage, MVVM with Clean Architecture, Retrofit/OkHttp, Room — delivering banking-grade reliability for sensitive financial transactions.
- Picture Book Locker: Android cabinet control system with electromagnetic locks, UV sterilization, sensors, face recognition + QR code authentication. Borrowing **client** app with <3s checkout time, 24/7 self-service.

Tech: Kotlin, Java, Android SDK, Spring Cloud, REST APIs, MySQL, Redis, Docker, CI/CD, WebSocket, NDK/JNI, MQTT, Zigbee, RFID

Senior Android Engineer | 2013 - 2018

- Enterprise Messaging Platform: architected Android **client** for 5,000+ DAU and 500K+ daily messages with sub-200ms latency. Built NDK/JNI TCP/UDP transport layer, WebSocket connection persistence, offline message sync, JPush backup push channel, and OEM ROM keep-alive strategies (Xiaomi/Huawei/OPPO). Migrated core IM from unstable in-house C++ to Easemob **cloud**, eliminating 90%+ defects. Built FastDFS distributed file storage (100GB+). Applied hotfix framework (Tinker, AndFix) for emergency patches and performance quality **engineering** across cold start, ANR, OOM, and APK size.
- Live Streaming Commerce System: full-**stack** Android + web delivery for live commerce platform. Android **client** in Java/Kotlin with RTMP/WebRTC/HLS streaming, multi-language architecture (C# + C++ media core + Python analytics + Lua config), 1000+ peak concurrency, 99.5% stream availability.

Tech: Kotlin, Java, Android SDK, NDK/JNI, C++, WebSocket, TCP/UDP, Gradle, CI/CD, Tinker, AndFix, JPush, RTMP, WebRTC, HLS, Ea

Additional Experience — Auckland, New Zealand

2025 – 2026

Customer Service & Operations (Part-time)

- Gilmours: handled inventory and order workflows while communicating with customers to resolve issues and keep service running smoothly
- Red Cross Shop: collaborated with volunteers and staff on handovers, shelf organization, and **customer** support

EDUCATION

Master of Computer and Information Sciences Auckland University of Technology, NZ 2024 – 2026

Research: Computer Vision, Diffusion Models, Large Language Models, Multimodal AI

Bachelor of Software Engineering Hebei Univ. of Sci. & Tech., China 2009 – 2013

Coursework: **Data** Structures, Algorithms, Java Programming, Database Systems, Software **Engineering**

LICENSES & CERTIFICATIONS

IoT Fundamentals — Xi'an Jiaotong University 2020

Science and Technology Achievement Award (Smart Manufacturing) — Hebei, China 2020

Software Design Engineer Certificate — China Qualification Authority 2019

INTERESTS

Hiking, self-hosted infrastructure, AI tools, and mobile technology trends

REFEREES

References available upon request.